

Product Documentation

CONVERGENCE-Ai™ Cloud-Native AI Automations Hub System Specifications

CONVERGENCE-Ai™ SYSTEMS CORPORATION

Vault-Grade Cryptographic Intelligence Systems

1. Executive Summary & Product Architecture

The CONVERGENCE-Ai™ Cloud-Native AI Automations Hub represents a next-generation middleware platform designed to bridge autonomous AI agent orchestrators (such as Dify.ai, LangGraph, and n8n) directly into active business web terminals. Constructed around a vault-grade multi-tenant architecture, the system isolates credentials, databases, and operational flows on a per-tenant basis, enabling secure deployment of AI workers across SMB networks, enterprise operations teams, and reseller channels.

2. The Vault-Grade Cryptographic Shield

A cornerstone of the platform is the Vault-Grade Cryptographic Shield. Utilizing AES-256-CBC envelope encryption, sensitive customer environment parameters, API tokens, and database credentials are fully encrypted before storage in the metadata database. Decryption is performed strictly at execution time using HSM keys managed under Google Cloud Key Management Service (Cloud KMS). Network architectures isolate administrative SSH endpoints exclusively via Identity-Aware Proxy (IAP) tunnels, locking down standard entry vectors.

3. Live LLM Gateway & Reseller Markups

CONVERGENCE-Ai supports direct API connections to public/private models with cost tracking. The Deployment Hub allows configuring LLM Providers (Gemini, OpenAI, Claude, or local Ollama), preferred model versions, and API keys. Resellers can define utility markups (e.g. 30%) to automatically calculate pass-through token costs for client billing. Administrators can use the 'AI Compose' action inside the HITL queue to review model drafts and estimated costs before releasing tasks.

4. Technical Design (SDD) & API Specs

The platform operates as an Express.js API gateway running in containerized environments. State persistence is managed dynamically using Supabase PostgreSQL databases with Row-Level Security (RLS) policies configured to isolate transactions based on verified tenant tokens.

Core REST API Gateway Routes:

```
GET /api/status - Checks gateway container health.
POST /api/deploy-agent - Provisions Cloud Run container and generates JWT license token.
POST /api/verify-token - Decrypts and validates JWT license configuration metadata.
POST /api/ai-compose - Generates high-fidelity AI drafts for HITL review.
GET /api/hitl - Pulls pending tasks in the Human-in-the-Loop review queue.
POST /api/hitl/action - Approves or rejects pending tasks, updating PostgreSQL logs.
GET /api/integrations/:provider/health - Runs diagnostic health audits measuring api response latency.
GET /api/integrate/:provider - Initiates OAuth2 handshake redirects to external partner portals.
```

5. E2E Testing & Verification Reports

The suite undergoes continuous testing via zero-dependency CLI test runners and visual E2E simulation consoles. These test assertions validate Base64 token compilation, process mapping state nodes, HITL override mechanics, and social campaign scheduler triggers. Performance benchmarks verify SLA targets are maintained under 5 seconds per transaction block.